

MANU PRAKASH BHAT

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EDUCATION

R. V. College of Engineering

B.E. in Information Science and Engineering — CGPA: 8.54

2022 – 2026

Bengaluru, India

MES PU College

Pre-University (Class XII) — 94.33%

2020 – 2022

Sirsi, India

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Node.js, FastAPI, SQLAlchemy, LangChain, Scikit-learn

Databases & Tools: PostgreSQL, SQLite, Git, GitHub, Docker, Kubernetes, Prometheus, GitHub Actions (CI/CD)

Cloud & Concepts: Azure, REST APIs, DevSecOps, Machine Learning, Retrieval-Augmented Generation (RAG), LLMs (GPT-4o), Database Design

Hardware: ESP32, Arduino, Raspberry Pi

EXPERIENCE

Honeywell Technologies

Jan 2026 – Jul 2026

Software Engineer Intern

Bengaluru, India

- **AI-Powered NHI Governance Platform** (*Python, LangChain, GPT-4o, React, TypeScript, Azure*): Built a DevSecOps platform that auto-discovers, risk-scores, and governs machine identities (tokens, keys, secrets, service principals, TLS certificates) across GitHub and Azure.
- Engineered a deterministic 0–100 risk-scoring engine and a 7-rule policy-as-code layer with optional GPT-4o explanations via a LangChain agent; automated audits through scheduled CI/CD workflows with SAST/SCA gates and Slack, Jira, and email alerts.
- **DevSecOps SRE & Engineering-Intelligence Platform** (*React, TypeScript, FastAPI, PostgreSQL, Prometheus, Nginx*): Built a full-stack platform monitoring 400+ self-hosted CI/CD runners across ~350 hosts via a runner-to-host metric-correlation engine.
- Added a stateful alerting engine, SSO with role-based access control (RBAC) across 7 roles, and “Unified Ops” — a config-driven pipeline unifying security, quality, and build metrics from 5 enterprise tools into one RBAC-scoped dashboard.
- **AI Dependency-Intelligence Assistant (RAG Chatbot)** (*FastAPI, Next.js, NetworkX, Azure OpenAI*): Built a hybrid Retrieval-Augmented Generation assistant answering plain-English questions over a 1,000+ package dependency graph for a large industrial software product.
- Grounded every answer in a deterministic NetworkX graph engine so the LLM phrases — but never hallucinates — impact-analysis facts; delivered a streaming (SSE) chat UI with interactive graph visualizations, secured by SSO/JWT.

PROJECTS

AI-Based Web Application Firewall | *Python, FastAPI, React, Scikit-learn*

2025

- Built an intelligent WAF combining ensemble ML and rule-based detection for SQL-injection, XSS, and command-injection attacks, reaching 95% detection accuracy.
- Designed a real-time inference pipeline with modular REST APIs, adaptive rule updates, rate limiting, IP allow/deny lists, and a live threat-analytics dashboard.

AI-Powered Sign Language to Speech Converter | *ESP32, FastAPI, React*

2025

- Designed an IoT glove (5 flex sensors + 6-axis IMU) with a Random Forest model reaching 92.4% accuracy at <40ms latency across all 26 ASL alphabets.
- Built power-efficient ESP32 firmware, a FastAPI inference service, and a React UI with real-time gesture recognition and text-to-speech synthesis.

ACHIEVEMENTS & CERTIFICATIONS

- **Honeywell STAR Award — “Drive Accountability Culture” (2026)**: recognized for exceptional ownership and execution in delivering a GenAI-based security and runner-monitoring solution advancing to production.
- **National Level INSPIRE Awards**: participant for innovative project development (2016).
- **Certifications**: Infosys Springboard — Operating System Fundamentals; Udemy — Mastering Data Structures & Algorithms using C/C++.